

Physics — Complete Test

Practice Questions with Answers and Explanations

Question: 141

Language: English

How to use this book

This collection contains all 141 questions of the test “Physics — Complete Test”. Every question shows all answer choices, the correct answer highlighted in green, and a detailed explanation below it. Try to answer on your own first, then check yourself. The answer key on the last page allows quick self-testing.

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1. The force $F(x,y,z)=axi+2byj-3czk$, where a,b and c are constant of suitable size, is conservative?

- A. Just when $a=-2b+3c$
- B. Always ✓
- C. Just when $a=2b-3c$
- D. Never

EXPLANATION

B, because a force is conservative if $\nabla \times F = 0$, which holds for $F = (ax, 2by, -3cz)$.

2. What is the gravity acceleration g present in a place where a simple pendulum of length 150.3 cm makes 100 oscillations in 246.7 s?

- A. $g = 9.949 \text{ m/s}^2$
- B. $g = 9.549 \text{ m/s}^2$
- C. $g = 9.449 \text{ m/s}^2$
- D. $g = 9.749 \text{ m/s}^2$ ✓

EXPLANATION

D, because g

3. In a region of space there is the potential energy $U(x, y, z) = b(x^2 + y^2 + z^2) - bxy + c$ where b and c are constants of suitable size. What is the analytical expression of the force?

- A. $F(x,y,z)=b(-2x+xy)i+b(-2y+xy)j-2bzk$
- B. $F(x,y,z)=b(2x-y)i+b(2y-x)j+(2bz+c)k$
- C. $F(x,y,z)=b(2x-y)i+b(2y-x)j+2bzk$
- D. $F(x,y,z)=b(-2x+y)i+b(-2y+x)j-2bzk$ ✓

EXPLANATION

Force is the negative gradient of potential energy, so $F = -\nabla U = -[2bx - by, 2by - bx, 2bz]$.

4. The radius of the Earth is about $R_E = 6370$ km. An object with mass 20 kg is taken to a height of 160 km above the Earth's surface. 1) What is the object's mass at this height? 2) How much does the object weigh at this height?

- A. 1) The mass decreases; 2) The weight is 196.0 N
- B. 1) The mass is the same; 2) The weight is 186.5 N ✓
- C. 1) The mass is the same; 2) The weight is 196.0 N
- D. 1) The mass decreases; 2) The weight is 186.5 N

EXPLANATION

1) Mass is constant (independent of height); 2) Weight = $mg' = 20 \text{ kg} \times 9.325 \text{ m/s}^2 = 186.5 \text{ N}$ ($g' = GM/(R_E + 160 \text{ km})^2$).

5. A point particle has a potential energy $U(x,y,z)=kx^2/2+hy^3+mgz$ where k and h are constants (J/m^2 and J/m^3), m is the particle mass and g is the gravitational acceleration. The motion equations are: a) $d^2x(t)/dt^2=-kx(t)/m$; $d^2y(t)/dt^2=-3hy^2(t)/m$; $d^2z(t)/dt^2=-g$ b) $d^2x(t)/dt^2=-kx(t)/m$; $d^2y(t)/dt^2=-3hy^2(t)/m$; $d^2z(t)/dt^2=-gz(t)$ c) $d^2x(t)/dt^2=kx(t)/m$; $d^2y(t)/dt^2=3hy^2(t)/m$; $d^2z(t)/dt^2=g$ d) $d^2x(t)/dt^2=-kx(t)/m$; $d^2y(t)/dt^2=-hy^2(t)/m$; $d^2z(t)/dt^2=g$

- A. $d^2x(t)/dt^2=-kx(t)/m$; $d^2y(t)/dt^2=-3hy^2(t)/m$; $d^2z(t)/dt^2=-g$ ✓
- B. $d^2x(t)/dt^2=-kx(t)/m$; $d^2y(t)/dt^2=-3hy^2(t)/m$; $d^2z(t)/dt^2=-gz(t)$
- C. $d^2x(t)/dt^2=kx(t)/m$; $d^2y(t)/dt^2=3hy^2(t)/m$; $d^2z(t)/dt^2=g$
- D. $d^2x(t)/dt^2=-kx(t)/m$; $d^2y(t)/dt^2=-hy^2(t)/m$; $d^2z(t)/dt^2=g$

EXPLANATION

Force = $-\nabla U \Rightarrow a_x = -kx/m$, $a_y = -3hy^2/m$, $a_z = -g$.

6. A physical pendulum swings around an axis passing through a point P located a distance d from the center of mass. The pendulum has:

- Mass M ,
- Moment of inertia I about the rotation axis,
- Distance d from the center of mass to axis P.

What is the period T of small oscillations?

- A. $T = 2\pi\sqrt{(I / Mgd)}$ ✓
- B. $T = 2\pi\sqrt{(Mgd / I)}$
- C. $T = 2\pi\sqrt{(Id / Mg)}$
- D. $T = 2\pi\sqrt{(d / g)}$

EXPLANATION

Period formula for physical pendulum is $T = 2\pi\sqrt{(I / Mgd)}$, matching option A.

7. In a region of space there is the potential energy $U(x, y, z) = b(x^2 + y^2 + z^2) - bxy + c$ where b and c are constants of suitable size. What is the analytical expression of the force?

- A. $F(x,y,z)=b(-2x+xy)i+b(-2y+xy)j-2bzk$
- B. $F(x,y,z)=b(2x-y)i+b(2y-x)j+(2bz+c)k$
- C. $F(x,y,z)=b(2x-y)i+b(2y-x)j+2bzk$
- D. $F(x,y,z)=b(-2x+y)i+b(-2y+x)j-2bzk$ ✓

EXPLANATION

Force is the negative gradient of potential energy, so $F = -\nabla U$.

8. Two satellites are placed in circular orbits of equal radius, one around the Earth and the other around the Moon. Which statement is correct?

- A. The terrestrial satellite makes one turn in a longer time than the lunar satellite
- B. The terrestrial satellite makes one revolution at the same time as the lunar satellite
- C. The terrestrial satellite makes one turn in a shorter time than the lunar satellite ✓
- D. To answer you must know the masses of the satellites

EXPLANATION

C, because Earth's stronger gravity increases orbital speed for the same radius.

9. In a region of space there is this potential energy $U(x,y,z) = x^2 - 3y^2 + xz$. What is the analytical expression of the force?

- A. $F(x,y,z)=-2i+6yj-zk$
- B. $F(x,y,z)=(-2x-z)i+6yj-xk$ ✓
- C. $F(x,y,z)=-2xi+6yj-xk$
- D. $F(x,y,z)=2xi-6yj+xk$

EXPLANATION

Force is the negative gradient of potential energy, so $F = -\nabla U = -(2x + z)i + 6yj - xk$.

10. To calculate the orbit of a planet around the Sun, in addition to the mass of the planet and the Sun, it is necessary and sufficient to know: (a) the radius of the planet and of the Sun (b) the total angular momentum of the system (c) the total energy and the total angular momentum of the system (d) the total energy of the system

- A. the radius of the planet and of the Sun
- B. the total angular momentum of the system
- C. the total energy and the total angular momentum of the system ✓
- D. the total energy of the system

EXPLANATION

C, because both energy and angular momentum are needed to fully describe the orbit.

11. In a region of space there is this potential energy $U(x,y,z)=x^2-3y^2+xz$. What is the analytical expression of the force?

- A. $F(x,y,z)=2xi-6yj+xk$
- B. $F(x,y,z)=-2i+6yj-zk$
- C. $F(x,y,z)=-2xi+6yj-xk$
- D. $F(x,y,z)=(-2x-z)i+6yj-xk$ ✓

EXPLANATION

Force is negative gradient of potential: $F = -\nabla U = -2xi + 6yj - xk$.

12. We imagine replacing the core of our planet with a material of such high density as to double the mass of the Earth, without altering its other quantities such as volume, shape, etc. What would happen to atmospheric pressure?

- A.** It would also double ✓
- B.** It would remain unchanged
- C.** It would be halved
- D.** It would become four times larger

EXPLANATION

Gravitational force (and thus pressure) scales with mass, so doubling Earth's mass doubles atmospheric pressure.

13. In a region of space there is this potential energy $U(x,y,z) = x^3 - y^3 + z$. What is the analytical expression of the force?

- A.** $F(x,y,z) = -3x^2\mathbf{i} + 3y^2\mathbf{j} - \mathbf{k}$ ✓
- B.** $F(x,y,z) = 3x^2\mathbf{i} - 3y^2\mathbf{j} + \mathbf{k}$
- C.** $F(x,y,z) = 3x^2\mathbf{i} + 3y^2\mathbf{j} + \mathbf{k}$
- D.** $F(x,y,z) = x^3\mathbf{i} - y^3\mathbf{j} + z\mathbf{k}$

EXPLANATION

Force is the negative gradient of potential energy: $\mathbf{F} = -\nabla U$.

14. The radius of the Moon is $R = 1740$ km and the acceleration of gravity at its surface is $g = 1.63$ m/s². What is the approximate speed of a spacecraft in circular orbit 10 km above the lunar ground?

- A.** 1800 m/s
- B.** 1680 m/s ✓
- C.** 1900 m/s
- D.** 1300 m/s

EXPLANATION

B, because orbital speed is calculated as $\sqrt{gR} \approx 1680$ m/s.

15. In a region of space there is this potential energy $U(x, y, z) = x^2 - 3y^2 + xz$. What is the analytical expression of the force?

- A.** $F(x,y,z) = -2\mathbf{i} + 6y\mathbf{j} - z\mathbf{k}$
- B.** $F(x,y,z) = (-2x - z)\mathbf{i} + 6y\mathbf{j} - x\mathbf{k}$ ✓
- C.** $F(x,y,z) = -2x\mathbf{i} + 6y\mathbf{j} - x\mathbf{k}$
- D.** $F(x,y,z) = 2x\mathbf{i} - 6y\mathbf{j} + x\mathbf{k}$

EXPLANATION

Force is the negative gradient of potential energy, so $\mathbf{F} = -\nabla U$.

16. How much is the gravitational field (F) and the gravitational potential energy (V) produced by a homogeneous spherical mass (M) of radius (R) at a point inside the sphere, at a distance ($r < R$) from the center? (Here, (C) is an integration constant.)

- A. ($F = -GMr/r^3$), ($V = GM/r^2 + C$)
- B. ($F = -GMr/R^3$), ($V = GMr^2/2R^3 + C$) ✓
- C. ($F = -GM/R^2$), ($V = GM/2R + C$)
- D. ($F = -GM/r^2$), ($V = GM/r + C$)

EXPLANATION

Inside a sphere, $F = -GMr/R^3$ and $V = GMr^2/2R^3 + C$, matching option B.

17. In a region of space there is the potential energy $U(x,y,z) = 2ax^2 - 3axy + az^2$ where a is a constant of suitable size. What is the analytical expression of the force?

- A. $F(x,y,z) = -4ax \mathbf{i} + 3ay \mathbf{j} - 2az \mathbf{k}$
- B. $F(x,y,z) = -4ax \mathbf{i} + 3ax \mathbf{j} - 2az \mathbf{k}$
- C. $F(x,y,z) = a(3y - 4x) \mathbf{i} + 3ax \mathbf{j} - 2az \mathbf{k}$ ✓
- D. $F(x,y,z) = 4ax \mathbf{i} - 3ay \mathbf{j} + 2az \mathbf{k}$

EXPLANATION

Force is the negative gradient of potential energy, so $F = -\nabla U$.

18. The gravitational force ($F = -GMm/r^2 \hat{u}_r$) acts between two spherical homogeneous masses (M) and (m) placed at a distance (r). Is this force conservative?

- A. No, because ($\nabla \times F < 0$)
- B. Yes, because ($\nabla \times F = 0$) ✓
- C. No, because ($\oint F \cdot d\vec{s} \neq 0$)
- D. No, because ($\nabla \times F \neq 0$)

EXPLANATION

A force is conservative if $\nabla \times F = 0$, which holds for gravitational force.

19. The speed of a meteor approaching the Earth is measured as $v_i = 100$ m/s at a distance of 900 km above sea level. At what speed $v_{(F)}$ does this meteor crash on the Earth's surface? (For the Earth: $M_{(E)} = 5.98 \times 10^{24}$ kg, $R_{(E)} = 6400$ km)

- A. $v_{(F)} = 278$ km/s
- B. $v_{(F)} = 2780$ km/s
- C. $v_{(F)} = 27.8$ km/s
- D. $v_{(F)} = 2.78$ km/s ✓

EXPLANATION

Use energy conservation: $\frac{1}{2}v_i^2 - G M_{(E)}/(R_{(E)} + 900\text{km}) = \frac{1}{2}v_{(F)}^2 - G M_{(E)}/R_{(E)} \rightarrow v_{(F)} \approx 2.78$ km/s.

20. Calculate the moment of inertia of a system consisting of the three masses shown in the figure with respect to each of the x , y and z axes.

- A. $I_x=2.0$ kg m²; $I_y=4.8$ kg m²; $I_z=6.8$ kg m² ✓
- B. $I_x=20$ kg m²; $I_y=4.8$ kg m²; $I_z=68$ kg m²
- C. $I_x=20$ kg m²; $I_y=48$ kg m²; $I_z=68$ kg m²
- D. $I_x=20$ kg m²; $I_y=48$ kg m²; $I_z=6.8$ kg m²

EXPLANATION

Moment of inertia is the sum of mass times distance squared for each axis: $I_x = 1(2^2) + 2(0^2) + 3(2^2) = 2$, $I_y = 1(1^2) + 2(2^2) + 3(1^2) = 4.8$, $I_z = 1(1^2) + 2(2^2) + 3(2^2) = 6.8$.

21. To know the force exerted by an object weighing 1000 kg on the floor of a room, you need to know:

- A. the area on which the body rests, the position of the center of gravity, and the density of the body
- B. nothing else ✓
- C. the area on which the body rests and the position of the center of gravity
- D. the area on which the body rests

EXPLANATION

Force = weight = mg (independent of area or position).

22. When the force $F(x,y,z)=2ayi-2bxj+9(a+b)yk$, where a and b are constants of suitable size, is conservative?

- A. (a) Always
- B. (b) When $a=-b$ ✓
- C. (c) When $a=b$
- D. (d) Never

EXPLANATION

Force is conservative if $(\partial F_x)/(\partial y) = (\partial F_y)/(\partial x)$, requiring $a = -b$.

23. The gravitational potential $V(r)$ of a homogeneous spherical mass M of radius R , at a point where $r > R$, is equal to: (here r is the distance from the center of the sphere)

- A. $V(r) = -\gamma M/r^2$
- B. $V(r) = -\gamma MR/r^2$
- C. $V(r) = -\gamma M/r$ ✓
- D. $V(r) = -\gamma M/R^2$

EXPLANATION

Gravitational potential outside a sphere is $V(r) = -\gamma M/r$, matching option C.

24. A sphere with radius $R=0.25$ m and mass $M=1$ Kg released on an inclined plane at height $h=10.5$ m from the ground, will the velocity of the center of mass be once it reaches the ground?

- A. (a) 4.0 m/s
- B. (b) 1.2 m/s
- C. (c) 24.0 m/s
- D. (d) 12 m/s ✓

EXPLANATION

D, because velocity is $\sqrt{2gh} \approx 12$ m/s (ignoring rotation).

25. How much is the gravitational force F and energy U exerted between two homogeneous spherical masses M_1 and M_2 , if the distance between their centers is r , with r greater than the radii of each mass?

- A. $F = -G M_1 M_2 r / r^3$; $U = G M_1 M_2 / r$
- B. $F = -G M_1 M_2 r / r^2$; $U = G M_1 M_2 / r^2$
- C. $F = -G M_1 M_2 / r^2$; $U = G M_1 M_2 / r$ ✓
- D. $F = -G M_1 M_2 r / r^3$; $U = G M_1 M_2 / r^2$

EXPLANATION

Gravitational force is $F = -G M_1 M_2 / r^2$ and potential energy is $U = -G M_1 M_2 / r$.

26. A planet describes an elliptical orbit around a star. Let $\omega = d\theta/dt$ its angular velocity and r its distance from the center of the star in a given instant. Is the following relationship correct?

- A. $\omega \cdot r = \text{constant}$
- B. $\omega/r = \text{constant}$
- C. $\omega/r^2 = \text{constant}$
- D. $\omega \cdot r^2 = \text{constant}$ ✓

EXPLANATION

Angular momentum conservation implies $\omega \cdot r^2 = \text{constant}$, so $\omega \cdot r = \text{constant}$.

27. What is the unit measurement of the universal gravitational constant G ?

- A. It is a pure number
- B. $\text{m}^3/(\text{kg} \cdot \text{s}^2)$ ✓
- C. $\text{m}^3/(\text{kg}^2 \cdot \text{s})$
- D. $\text{m}^2/(\text{kg}^2 \cdot \text{s}^2)$

EXPLANATION

G has units of $\text{m}^3/(\text{kg} \cdot \text{s}^2)$ from Newton's law of gravitation.

28. A block of mass $m_1 = 1 \text{ kg}$ is attached to one end of a rope going around a pulley. Another block of mass $m_2 = 2 \text{ kg}$ at rest on a frictionless horizontal plane is tied to the other end of the rope. The pulley has a mass $m_2 = 2 \text{ kg}$, a radius $R = 20 \text{ cm}$ and a moment of inertia $I = 3 \text{ kgm}^2$. Calculate the tensions in the ropes.

- A. $T_1 = 0.9 \text{ N}$; $T_2 = 0.3 \text{ N}$
- B. $T_1 = 9.9 \text{ N}$; $T_2 = 9.3 \text{ N}$
- C. $T_1 = 9.9 \text{ N}$; $T_2 = 3.3 \text{ N}$
- D. $T_1 = 9.9 \text{ N}$; $T_2 = 0.3 \text{ N}$ ✓

EXPLANATION

Use Newton's second law and torque equation: $T_1 - m_1 g = m_1 a$, $T_2 = m_2 a$, $(T_1 - T_2)R = I\alpha \rightarrow T_1 = 9.9 \text{ N}$, $T_2 = 0.3 \text{ N}$.

29. Two liters of water at temperature $T_1 = 360 \text{ K}$ are mixed with four liters of water at temperature $T_2 = 320 \text{ K}$ in a thermally insulated container. The specific heat of water is $c = 4186 \text{ J (Kg K)}$. The change in entropy of the system is

- A. Negative
- B. There are insufficient data to answer the question
- C. Positive ✓
- D. Nothing

EXPLANATION

Entropy increases in irreversible mixing of two water masses at different temperatures.

30. A mass $m=1$ kg is attached to the end of a rope that is wrapped around a pulley with radius $R=20$ cm and with moment of inertia $I=0.5$ kgm². The mass is released at a height of $h=2.7$ m from the ground. With what velocity v will it hit the ground?

- A. $v=2$ m/s. ✓
- B. $v=0.2$ m/s.
- C. $v=20$ m/s.
- D. $v=12$ m/s.

EXPLANATION

Energy conservation: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$, solve for v .

31. Which of the following second law statements is incorrect?

- A. The entropy of a hot copper block decreases as it cools
- B. Ice is melted in water in an insulated container, the net entropy decreases ✓
- C. The entropy of an isolated systems must remain constant or increase.
- D. Work must be input & energy is transferred from a cold body to a hot body

EXPLANATION

Entropy of an isolated system **never decreases** (2nd law).

32. A rod with mass m and length $L=1.2$ m is hinged from one end to a wall. The rod is released in horizontal position (1). What will its angular velocity ω be once it reaches vertical position (2)?

- A. $\omega=5$ s⁻¹ ✓
- B. $\omega=15$ s⁻¹
- C. $\omega=0.5$ s⁻¹
- D. $\omega=0.25$ s⁻¹

EXPLANATION

Angular velocity $\omega = \sqrt{(3g/L)} = \sqrt{((3 \times 9.8)/1.2)} \approx 5$ s⁻¹.

33. Which of the following statements can be taken as the definition of reversible transformation?

- A. It is a transformation in which friction phenomena are absent
- B. It is a transformation that passes through a succession of states of equilibrium ✓
- C. It is a cyclic transformation
- D. It is a transformation in which entropy remains constant

EXPLANATION

Reversible transformations pass through equilibrium states.

34. The angular momentum of a rigid body revolving around an axis is doubled. How does the rotational kinetic energy of the body vary?

- A. It becomes 16 times greater
- B. It becomes 4 times greater ✓
- C. It doubles
- D. It becomes 8 times greater

EXPLANATION

Kinetic energy $K = L^2/(2I)$, doubling L quadruples K .

35. Two liters of water at temperature $T_1 = 340$ K are mixed with four liters of water at temperature $T_2 = 300$ K in a thermally insulated container. The specific heat of water is $c = 4186$ J (Kg K). The change in entropy of the system is

- A. positive ✓
- B. negative
- C. nothing
- D. there is insufficient data to answer the question

EXPLANATION

Entropy increases in irreversible mixing of different-temperature fluids ($\Delta S = mc \ln((T_{\text{f}})/T_1) + mc \ln((T_{\text{f}})/T_2) > 0$).

36. A motorcycle showman with mass m drives through a vertical circular track with radius $r = 40$ m. What should his minimum speed v be at the top point so that the motorcycle will not lose contact with the track?

- A. $v = 20$ m/s ✓
- B. $v = 40$ m/s
- C. $v = 10$ m/s
- D. $v = 30$ m/s

EXPLANATION

At the top, centripetal force equals gravity: $v = \sqrt{g r} = \sqrt{9.8 \times 40} \approx 20$ m/s.

37. Two kilograms of air is heated at constant pressure of $2 \cdot 10^5$ Pa to 500 °C. Calculate the entropy change ΔS if the initial volume is 0.8 m³ (air molecular weight 28.96 g/mol)

- A. $\Delta S = 2040$ J/K ✓
- B. $\Delta S = 20.40$ J/K
- C. $\Delta S = 2.040$ J/K
- D. $\Delta S = 204.0$ J/K

EXPLANATION

$\Delta S = nR \ln(V_2/V_1) + nC_p \ln(T_2/T_1) \approx 2040 \text{ J/K}$

38. A solid sphere with mass M and radius R is welded to the end of a rod with mass m and length L . Calculate the moment of inertia of this system with respect to the y -axis on the other end.

- A. $I_y = 1/12 mL^2 + 2/5 MR^2$
- B. $I_y = 1/3 mL^2 + 2/5 MR^2$
- C. $I_y = 1/3 mL^2 + 2/5 MR^2 + M(L+R)^2$ ✓
- D. $I_y = M(L+R)^2$

EXPLANATION

$I_y = I_{\text{rod}} + I_{\text{sphere}} + Md^2 = 1/3 mL^2 + 2/5 MR^2 + M(L+R)^2$.

39. The Internal energy U and entropy S functions have what in common: (a) both depend only on the temperature. (b) both do not depend on the temperature, their variations are always greater than zero (c) they are both function of the state of a thermodynamic system. (d) both do not depend on the temperature, their variations are always greater than zero

- A. both depend only on the temperature.
- B. both do not depend on the temperature, their variations are always greater than zero
- C. they are both function of the state of a thermodynamic system. ✓
- D. both do not depend on the temperature, their variations are always greater than zero

EXPLANATION

Both internal energy (U) and entropy (S) are state functions, depending only on the system's state.

40. A pencil of length $L = 15$ cm and mass m is placed in a vertical position on a plane with friction. Initially stopped, it falls by rotating around the point of contact with the plane. Calculate angular speed ω at the impact instant with the plane

- A. $\omega = 14 \text{ s}^{-1}$ ✓
- B. I cannot answer because I don't know the pencil mass
- C. $\omega = 1.40 \text{ s}^{-1}$
- D. $\omega = 0.14 \text{ s}^{-1}$

EXPLANATION

$\omega = \sqrt{(3g/L)} \approx 14 \text{ s}^{-1}$, independent of mass.

41. A thermal machine operates with three heat sources T_1 , T_2 , and T_3 are the temperatures of the sources and Q_1 , Q_2 , and Q_3 are amount of heat that the thermal machine exchanges with each of the sources. If the cycle of the thermal machine is irreversible. Then it is always true that

- A. $Q_1/T_1 + Q_2/T_2 + Q_3/T_3 < 0$ ✓
- B. $|Q_1|/T_1 + |Q_2|/T_2 + |Q_3|/T_3 = 0$
- C. $Q_1/T_1 + Q_2/T_2 + Q_3/T_3 > 0$
- D. $Q_1/T_1 + Q_2/T_2 + Q_3/T_3 = 0$

EXPLANATION

For irreversible cycles, the Clausius inequality states $Q_1/T_1 + Q_2/T_2 + Q_3/T_3 > 0$.

42. A homogeneous solid cylinder of length $l = 40$ cm, radius $r = 2$ cm and a mass $m = 2$ Kg rotates in a horizontal plane around a vertical axis perpendicular to its axis in the middle point O as a result of an impulse $J = 5 \times 10^{-2}$ Ns received perpendicular to an extreme. As a result of the friction the cylinder stops after traveling a lap. Calculate the initial angular speed ω_0 of the cylinder

- A. $\omega_0 = 0.37 \text{ s}^{-1}$ ✓
- B. $\omega_0 = 3.70 \text{ s}^{-1}$
- C. $\omega_0 = 0.07 \text{ s}^{-1}$
- D. $\omega_0 = 37.00 \text{ s}^{-1}$

EXPLANATION

Angular momentum $J \cdot R = I\omega_0 \rightarrow \omega_0 = J \cdot R / (\frac{1}{2}mR^2) = 0.37 \text{ s}^{-1}$.

43. The Poisson law for the gravitational potential V is (where ρ is the mass density and G the gravitational constant):

- A. $(\partial^2 V)/(\partial x^2) + (\partial^2 V)/(\partial y^2) + (\partial^2 V)/(\partial z^2) = \rho$
- B. $(\partial V)/(\partial x) + (\partial V)/(\partial y) + (\partial V)/(\partial z) = 4\pi G\rho$
- C. $(\partial^2 V)/(\partial x^2) + (\partial^2 V)/(\partial y^2) + (\partial^2 V)/(\partial z^2) = -4\pi G\rho$ ✓
- D. $(\partial V)/(\partial x) + (\partial V)/(\partial y) + (\partial V)/(\partial z) = -4\pi G\rho$

EXPLANATION

Poisson equation for gravitational potential is $\nabla^2 V = -4\pi G\rho$.

44. The force $F(xyz) = a y \text{ extbf{i}} + b x \text{ extbf{j}} + 3 \text{ extbf{k}}$, where a and b are constant of suitable size, is conservative?

- A. (a) Never
- B. (b) Just when $a = b$ ✓
- C. (c) Just when $a = -b$
- D. (d) Always

EXPLANATION

A vector field is conservative if $(\partial F_x)/(\partial y) = (\partial F_y)/(\partial x) \rightarrow a = b$.

45. An inventor claims that a thermal engine, operating between ocean layers at 27°C and 10°C , produces 10 kW of power while discharging 9900 kJ/min. This engine is:

- A. impossible ✓
- B. reversible
- C. possible
- D. probable

EXPLANATION

Efficiency $\eta = 1 - T_{\text{Cold}}/T_{\text{hot}} = 1 - 283/300 \approx 0.057 < 10/9900 \text{ min}^{-1}$.

46. A wheel of radius R rotates around a fixed axis. When a point, at a distance R from the center, moves with an angular speed ω , a second point, located at a distance $R/2$ from the center, moves with an angular velocity

- A. ω ✓
- B. 2ω
- C. $\omega/2$
- D. 4ω

EXPLANATION

Angular velocity is independent of radial distance, Q2: $\Delta S_{\text{total}} = \Delta S_{\text{ice}} + \Delta S_{\text{water}} = (10 \cdot 340/273) + (20 \cdot 4.186 \cdot \ln(293/273)) \approx 0.21 \text{ kJ/K}$, Q3: $W = \int \mathbf{F} \cdot d\mathbf{r} = k \int (xi+yj+zk) \cdot (dx+dy+dz)$ from (0,0,0) to (1,0,1) = $2(1+0+1)/2 = 2\text{J}$, Q4: $\Delta S = Q/T = (Mgh)/T = (10 \cdot 9.8 \cdot 10)/293 \approx 3.3 \text{ J/K}$, Q5: $\eta = 1 - T_c/T_h = 1 - 293/473 \approx 0.38$, $Q_{\text{out}} = Q_{\text{in}} - W = 10/0.38 - 10 \approx 16.3 \text{ kJ/s}$, Q6: $\text{COP} = Q_c/W = T_c/(T_h - T_c) \rightarrow T_h = T_c + Q_c/W = 293 + 20/10 = 313\text{K} \rightarrow 440\text{K}$, Q7: $PV^\gamma = \text{const}$ and $TV^{(\gamma-1)} = \text{const}$ are adiabatic for ideal gas, Q8: $Q = n \int C_v dT = 2 \int (C_p - R) dT = 2 \int (2 + 0.02T - 8.314) dT$ from 400 to 800 = $8.65 \cdot 10^3 \text{ J}$, Q9: The two laws contradict each other unless $\gamma=1$ (not possible for ideal gas), Q10: $W = nRT \ln(P_2/P_1) = (2000/28.96)(8.314)(293) \ln(600/100) \approx -220 \text{ kJ}$, Q11: $T \propto 1/\sqrt{g}$, $g \propto M \rightarrow T = 1/\sqrt{2} \text{ sec}$, Q12: $W = \int \mathbf{F} \cdot d\mathbf{r} = \int (2i + xyj + xz^2k) \cdot (0+0+3dz) = \int 3xz^2 dz$ from 1 to 4 = $3 \cdot 2 \cdot (16-1)/3 = 42\text{J}$, Q13: $W = nC_v \Delta T = (10/28.96)(8.314)(500-273) \approx 2990 \text{ kJ}$, Q14: $E = (l/2)k_B T$, Q15: $T_F = T_i(P_F/P_i)^{((\gamma-1)/\gamma)} = 300 \cdot 2^{(0.4/1.4)} \approx 361\text{K}$, $V_F = nRT_F/P_F \approx 3.0 \cdot 10^{-3} \text{ m}^3$, $W = nC_v(T_F - T_i) \approx -254\text{J}$, Q16: $V_b = V_a(P_a/P_b) = 25\text{L}$, $W = P_b(V_c - V_b) = 2 \cdot (25-10) = 30 \text{ atm} \cdot \text{L} \approx 3000\text{J}$, Q17: $T_f = T_i(P_f/P_i)^{((\gamma-1)/\gamma)} = 300 \cdot 3^{(0.4/1.4)} \approx 411\text{K}$, Q18: $\nabla \times \mathbf{F} = (0,0,0)$ always $\rightarrow$ conservative, Q19: $T = 2\pi v/(L/g) \rightarrow g = 4\pi^2 L/T^2 \approx 9.749 \text{ m/s}^2$, Q20: $\mathbf{F} = -\nabla U = b(-2x+y)\mathbf{i} + b(-2y+x)\mathbf{j} - 2bz\mathbf{k}$, Q21: Mass is invariant; $g' = GM'/(R+h)^2 \approx 9.51 \text{ m/s}^2 \rightarrow \text{Weight} = mg' \approx 186.5\text{N}$, Q22: $\mathbf{F} = -\nabla U \rightarrow a)$ correct, Q23: $T = 2\pi v/(l/Mgd)$, Q24: $T \propto 1/\sqrt{g}$, $g_{\text{earth}} > g_{\text{moon}} \rightarrow T_{\text{earth}} < T_{\text{moon}}$, Q25: $\mathbf{F} = -\nabla U = (-2x-z)\mathbf{i} + 6y\mathbf{j} - x\mathbf{k}$

47. Calculate the total entropy change if 10 kg of ice at 0C is mixed in an insulated container with 20 kg of water at 20 C. (The heat of melting for ice is 340 kJ/Kg).

- A. 6.1 kJ/K
- B. 3.9 kJ/K
- C. 1.2 kJ/K
- D. 0.21 kJ/K ✓

EXPLANATION

D, because the entropy change is minimal due to the small temperature difference and large heat capacity of water.

48. A block of mass $m_1 = 1 \text{ kg}$ is attached to one end of a rope going around a pulley. Another block of mass $m_2 = 2 \text{ kg}$ at rest on a frictionless horizontal plane is tied to the other end of the rope. The pulley has a mass $M = 2 \text{ kg}$, a radius $R = 20 \text{ cm}$ and a moment of inertia $I = 3 \text{ kg m}^2$. Calculate the linear acceleration a and angular accelerations α .

- A. $a=0.33 \text{ m/s}^2$; $\alpha=1.64 \text{ s}^2$
- B. $a=0.53 \text{ m/s}^2$; $\alpha=2.64 \text{ s}^2$
- C. $a=0.13 \text{ m/s}^2$; $\alpha=0.64 \text{ s}^2$ ✓
- D. $a=1.13 \text{ m/s}^2$; $\alpha=0.64 \text{ s}^2$

EXPLANATION

C, because $a = (m_2 g)/(m_1 + m_2 + I/R^2) \approx 0.13 \text{ m/s}^2$, $\alpha = a/R \approx 0.64 \text{ rad/s}^2$.

49. A system performs a thermodynamic cycle. Its entropy variation

- A. is always positive
- B. is null if the cycle is reversible
- C. depends on the thermodynamic cycle it is positive or negative
- D. is null ✓

EXPLANATION

Entropy change over a reversible cycle is zero ($\Delta S = \oint dQ/T = 0$).

50. A bullet with mass $m = 50 \text{ g}$ travels at a speed of $v = 200 \text{ m/s}$ and embeds itself into the rim of a disk at rest that can rotate freely, at a distance of $R = 60 \text{ cm}$ from the center. The disk mass is $M = 0.9 \text{ Kg}$. What is the angular speed of the system ($M + m$) after the collision?

- A. $\omega = 33 \text{ s}^{-1}$ ✓
- B. $\omega = 3.3 \text{ s}^{-1}$
- C. $\omega = 0.3 \text{ s}^{-1}$
- D. $\omega = 6.0 \text{ s}^{-1}$

EXPLANATION

Angular momentum conservation: $m \cdot v \cdot R = (I_{\text{disk}} + m \cdot R^2) \omega \rightarrow \omega \approx 33 \text{ s}^{-1}$.

51. A piston allows air to expand from $6 \cdot 10^6 \text{ Pa}$ to $2 \cdot 10^5 \text{ Pa}$. The initial volume and temperature are 500 cm^3 and 800 C . If the temperature is held constant, calculate the heat transfer Q and the entropy change ΔS (air molecular weight 28.96 g/mole).

- A. $Q = 1.02 \cdot 10^4 \text{ J}$; $\Delta S = 9.51 \text{ J/K}$ ✓
- B. $Q = 1.02 \cdot 10^3 \text{ J}$; $\Delta S = 9.51 \text{ J/K}$
- C. $Q = 1.02 \cdot 10^4 \text{ J}$; $\Delta S = 95.1 \text{ J/K}$
- D. $Q = 1.02 \cdot 10^3 \text{ J}$; $\Delta S = 95.1 \text{ J/K}$

EXPLANATION

$Q = nRT \ln(P_1/P_2) \approx 1.02 \times 10^4 \text{ J}$; $\Delta S = Q/T \approx 9.51 \text{ J/K}$.

52. A perfect gas thermal machine describes a cycle formed by a reversible isothermal expansion, a reversible isochoric transformation, an irreversible isothermal compression and a reversible isochoric transformation in a cycle (A) the entropy variation of the thermodynamic universe is zero (B) the entropy variation of the thermodynamic universe is negative impossible (C) the entropy variation of the gas is positive (D) the entropy variation of the gas is zero.

- A. the entropy variation of the thermodynamic universe is zero
- B. the entropy variation of the thermodynamic universe is negative impossible
- C. the entropy variation of the gas is positive
- D. the entropy variation of the gas is zero ✓

EXPLANATION

For a reversible cycle, the entropy change of the gas over one full cycle is zero.

53. A steel ball is released from a height h , falls to the ground and bounces back. Neglecting the friction with air, the height it will attain after the rebound

- A. 1. is equal to h only if the collision with the ground is perfectly elastic. ✓
- B. 2. is equal to h if the momentum is conserved during the collision with the ground.
- C. 3. is equal to h .
- D. 4. is equal to $h/2$.

EXPLANATION

Height after rebound equals initial height only if collision is perfectly elastic.

54. A mole of ideal gas is in the initial state P_0 , V_0 and T_0 . The gas is heated to a constant volume up to the temperature $2T_0$, then left to expand isothermally up to volume $2V_0$ and finally cooled in an isobaric way up to the temperature T_0 . The change of entropy is

- A. $\Delta S = (3 \cdot 8.31)/2 \cdot \ln(2)$ J/K
- B. $\Delta S = (5 \cdot 8.31)/2 \cdot \ln(2)$ J/K
- C. $\Delta S = (5 \cdot 8.31)/2$ J/K
- D. $\Delta S = 0$ J/K ✓

EXPLANATION

$\Delta S = 0$ because the process is cyclic, returning to the initial state, so the net entropy change is zero.

55. Two bullets of different mass are fired at the same speed with the same angle and from the same point. Neglecting the air resistance and assuming the ground surface is perfectly flat, which of the two takes the longest time to reach the ground?

- A. the data provided is insufficient to answer
- B. the one that has less mass
- C. they will both take the same time ✓
- D. the one with the greatest mass

EXPLANATION

Time of flight depends only on vertical motion, same for both bullets.

56. A mole of an ideal monoatomic gas travels along the cycle shown in the figure. To calculate the variations of internal energy and entropy in a complete cycle.

- A. $\Delta U = 3p_0V_0$ J; $\Delta S = -3p_0V_0/T_0$ J/K
- B. $\Delta U = 3p_0V_0$ J; $\Delta S = 3p_0V_0/T_0$ J/K
- C. $\Delta U = -3p_0V_0$ J; $\Delta S = 0$ J/K
- D. $\Delta U = 0$ J; $\Delta S = 0$ J/K ✓

EXPLANATION

$\Delta U = 0$ (internal energy is a state function), $\Delta S = 0$ (reversible cycle).

57. A block initially sliding on a rough horizontal surface with velocity of 10 m/s stops after travelling a distance of 70 m. Find the coefficient of dynamic friction μ_d between surface and the block.

- A. $\mu_d = 0.073$ ✓
- B. $\mu_d = 0.173$
- C. $\mu_d = 0.273$
- D. $\mu_d = 0.003$

EXPLANATION

$\mu_d = v^2/(2gd) = 100/(2 \cdot 9.8 \cdot 70) \approx 0.073$.

58. A copper block of mass $M = 0.5 \text{ kg}$ falls from a height of $h = 100 \text{ m}$ in a lake at temperature $T_{\text{(L)}} = 283 \text{ K}$. The initial temperature of the copper block is $T_{\text{Cu}} = 423 \text{ K}$. Calculate the change in entropy of the universe ΔS_{U} in this process. The specific heat of copper is $c = 387 \text{ J/KgK}$.

- A. $\Delta S_{\text{(U)}} = 9.71 \text{ J/K}$
- B. $\Delta S_{\text{(U)}} = 140.00 \text{ J/K}$
- C. $\Delta S_{\text{(U)}} = -9.71 \text{ J/K}$
- D. $\Delta S_{\text{(U)}} = 19.7 \text{ J/K} \checkmark$

EXPLANATION

$\Delta S_{\text{U}} = \Delta S_{\text{Cu}} + \Delta S_{\text{Lake}} = Mc \ln((T_{\text{(f)}})/T_{\text{i}}) + Q/(T_{\text{(L)}})$, where $Q = Mc\Delta T \rightarrow \Delta S_{\text{U}} = 19.7 \text{ J/K}$.

59. During the laying of a high voltage pylon, a bolt falls from a height h above the ground. It is known that in the last second of its motion before hitting the ground, it travels a height equal to $h/2$. Determine the value of h .

- A. $h = 37.2 \text{ m}$
- B. $h = 47.2 \text{ m}$
- C. $h = 67.2 \text{ m}$
- D. $h = 57.2 \text{ m} \checkmark$

EXPLANATION

Use kinematic equations for free fall, solve for h .

60. A block of 0.1 kg of ice at $T = 0^\circ \text{C}$ is completely melted always at temperature $T = 0^\circ \text{C}$. What is its variation of entropy ΔS (The latent heat of water is $3.335 \cdot 10^5 \text{ J/Kg}$)?

- A. $\Delta S = 0 \text{ J/K}$ because this thermal transformation is reversible
- B. $\Delta S = 122 \text{ J/K} \checkmark$
- C. $\Delta S = 1220$
- D. $\Delta S = -122 \text{ J/K}$

EXPLANATION

$\Delta S = m\lambda/T = 0.1 \times 3.335 \times 10^5 / 273 \approx 122 \text{ J/K}$.

61. A ballistic pendulum is a device used to calculate the muzzle velocities of bullets. It is a pendulum constructed by hanging a wooden block with a very high mass at the end of a rope with length L . When a bullet with mass m traveling at speed v embeds itself into the wooden block with mass M , the (bullet + block) system rises by height h . Calculate the speed of the bullet when: $m = 50 \text{ g}$, $M = 5 \text{ kg}$, $L = 2 \text{ m}$, $h = 80 \text{ cm}$

- A. $v = 4000 \text{ m/s}$
- B. $v = 4 \text{ m/s}$
- C. $v = 40 \text{ m/s}$
- D. $v = 400 \text{ m/s} \checkmark$

EXPLANATION

Conservation of momentum: $mv = (m+M)\sqrt{2gh} \rightarrow v \approx 400 \text{ m/s}$.

62. A body of mass m and specific heat c , initially at temperature T_i , exchanges heat from a single hot reservoir (at constant temperature T_h). The body is not a reservoir, so it has a finite heat capacity, and its temperature will increase up to T_h . What is the entropy variation of the Universe (system+environment)?

- A. $\Delta S_{(U)} = 0$
- B. $\Delta S_{(U)} = (mc(T_h - T_i))/T_h + mc \ln(T_h/T_i)$
- C. $\Delta S_{(U)} = (mc(T_i - T_h))/T_h + mc \ln(T_i/T_h)$ ✓
- D. Option D

EXPLANATION

$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{reservoir}} = mc \ln(T_h/T_i) - mc(T_h - T_i)/T_h$.

63. A ballistic pendulum consists of a large block of wood of mass $M = 9.99$ kg. A bullet of mass $m = 10$ g, travelling horizontally with speed v , hits the pendulum and remains stuck in it. After the collision, the pendulum oscillates and reaches a maximum height $h = 5$ cm. What is the velocity v of the bullet?

- A. $v = 5 \times 10^4$ m/s
- B. $v = 10^2$ m/s
- C. $v = 10^3$ m/s ✓
- D. $v = 10^4$ m/s

EXPLANATION

Conservation of momentum and energy: $v = ((M+m)/m)\sqrt{2gh} \approx 1000$ m/s.

64. Knowing that the mass of a planet is $M = 5.98 \times 10^{26}$ kg and the radius is $R = 6.37 \times 10^6$ m, calculate the escape velocity v_e of a body from the planet.

- A. $v_e = 1.11 \times 10^3$ m/s
- B. $v_e = 1.11 \times 10^4$ m/s
- C. $v_e = 1.11 \times 10^5$ m/s ✓
- D. It is not possible to answer because the mass of the body is not known

EXPLANATION

Escape velocity formula: $v_e = \sqrt{2GM/R} \rightarrow v_e = 1.11 \times 10^5$ m/s.

65. The force $F = k(x_i + y_j + z_k)$ acts on a point particle of mass $m = 2$ kg. What work does the force do to move the mass from the origin of the coordinates to point A(1,0,1) knowing that $k = 2$ N/m and the values of the coordinates of point A are in meters?

- A. 2J ✓
- B. -2J
- C. 0.4J
- D. -0.4J

EXPLANATION

A, because the work done by a conservative force is path-independent: $W = \int F \cdot dr = k(x^2 + y^2 + z^2)/2$ from (0,0,0) to (1,0,1) = $2(1+0+1)/2 = 2$ J.

66. A body of mass $M = 10 \text{ kg}$ falls from a height $h = 10 \text{ m}$ and stops. Initially the body and the environment are at the same temperature $T = 293 \text{ K}$. What is the entropy variation ΔS of the universe?

- A. $\Delta S = 33.3 \text{ J/K}$
- B. $\Delta S = -3.3 \text{ J/K}$
- C. $\Delta S = 0 \text{ J/K}$
- D. $\Delta S = 3.3 \text{ J/K}$ ✓

EXPLANATION

D, because the entropy increase from the irreversible process (falling mass) dominates: $\Delta S = Mgh/T = (10)(9.8)(10)/293 \approx 3.3 \text{ J/K}$.

67. If $a = -kv^2$, where k is a constant, determine $v(t)$ and $x(t)$ given the initial conditions: $x(0) = 0$ and $v(0) = v_0$

- A. $x(t) = (\ln(v_{0kt} - 1))/k$, $v(t) = v_0/(v_{0kt} + 1)$
- B. $x(t) = (\ln(v_{0kt} + 1))/k$, $v(t) = v_0/(v_{0kt} + 1)$ ✓
- C. $x(t) = (\ln(v_{0kt} + 1))/k$, $v(t) = v_0/(v_{0kt} + 1)$
- D. $x(t) = (e^{-kt})/k - t + 1/k$, $v(t) = e^{-kt} - 1$

EXPLANATION

Solve $a = -kv^2$ to get $v(t) = v_0/(v_{0kt} + 1)$ and integrate to find $x(t) = (\ln(v_{0kt} + 1))/k$.

68. A block of 0.1 kg of ice at $T = 0^\circ \text{C}$ is completely melted always at temperature $T = 0^\circ \text{C}$. What is its variation of entropy ΔS (The latent heat of water is $3.335 \cdot 10^5 \text{ J/Kg}$)

- A. $\Delta S = 0 \text{ J/K}$ because this thermal transformation is reversible
- B. $\Delta S = 122 \text{ J/K}$ ✓
- C. $\Delta S = 1220 \text{ J/K}$
- D. $\Delta S = -122 \text{ J/K}$

EXPLANATION

Entropy change $\Delta S = Q/T = (m \cdot L)/T$.

69. A particle moves along the x axis according to the equation $x = 4t^2$, where x is in meters and t is in seconds. Therefore:

- A. 1. the acceleration of the particle is 8 m/s^2 ✓
- B. 2. the particle follows a parabolic path
- C. 3. the speed is constant
- D. 4. cannot be negative

EXPLANATION

Acceleration $a = d^2x/dt^2 = 8 \text{ m/s}^2$, constant.

70. What is the efficiency η of a Stirling cycle that works between the temperature T_1 and $T_2 > T_1$?

- A. $\eta = 1 - T_1/T_2$ ✓
- B. $\eta = 1 + T_1/T_2$
- C. $\eta = 1 - T_2/T_1$
- D. $\eta = T_2 - T_1$

EXPLANATION

Stirling cycle efficiency $\eta = 1 - T_1/T_2$.

71. If $a = -k r^2$ where k is a constant, determine $v=v(t)$ and $x=x(t)$ knowing that $x(t=0)=0$ and $v(t=0)=v_0$

- A. $x(t)=(b(v_{0kt} - 1))/k$, $v(t)=v_0/((v_{0kt} + 1))$
- B. $x(t)=(b(v_{0kt} + 1))/k$, $v(t)=v_0/((v_{0kt} + 1))$ ✓
- C. $x(t)=(b(v_{0kt} + 1))/k$, $v(t)=v_0/((v_{0kt} + 1))$
- D. $x(t)=(\exp(-kt))/k - t + 1/k$, $v(t)=\exp(-kt) - 1$

EXPLANATION

Solve $dv/dt = -k \cdot r^2 = -k \cdot x^2 \rightarrow v(t) = v_0/((v_{0kt} + 1))$, then integrate to get $x(t)$.

72. What is the efficiency η of a Carnot cycle that works between the temperature $T_1 = 20^\circ\text{C}$ and $T_2 = 100^\circ\text{C}$?

- A. $\eta = 0.2144$ ✓
- B. $\eta = 4/5$
- C. $\eta = 1/5$
- D. $\eta = 0.7856$

EXPLANATION

Carnot efficiency $\eta = 1 - T_1/T_2 = 1 - 293/373 \approx 0.2144$.

73. The energy input to an engine is 3.00 times greater than the work it performs. What is its thermal efficiency?

- A. 3.00
- B. 1.00
- C. 0.333 ✓
- D. impossible to determine

EXPLANATION

Efficiency $\eta = W/Q_{\text{in}} = 1/3 \approx 0.333$.

74. The position of an object varies as $x(t) = A t^3$ (in meters), where $A = 1 \text{ m/s}^2$. What is its acceleration at time $t = 1 \text{ s}$, in m/s^2 ?

- A. 1
- B. 0
- C. 3
- D. 6 ✓

EXPLANATION

$a(t) = d^2x/dt^2 = 6A t$, so at $t = 1 \text{ s}$, $a = 6 \text{ m/s}^2$.

75. The entropy change ΔS of a body of mass M that changes phase at temperature T with latent heat λ is

- A. $\Delta S = M\lambda/T$ ✓
- B. $\Delta S = 0$
- C. $\Delta S = M\lambda T$
- D. $\Delta S = MT/\lambda$

EXPLANATION

Entropy change $\Delta S = M\lambda/T$ for phase change.

76. If $a = -kv$ where k is a constant, determine $v = v(t)$ and $x = x(t)$ knowing that $x(t=0) = 0$ and $v(t=0) = 0$

A. $x(t) = -(\exp(-kt))/k - t + 1/k$; $v(t) = \exp(-kt) - 1$ ✓

B. $x(t) = -\exp(-kt) - t + 1$; $v(t) = \exp(-kt) - 1$

C. $x(t) = -(\exp(-kt))/k - t$; $v(t) = \exp(-kt)$

D. $x(t) = -(\exp(-kt))/k - t$; $v(t) = \exp(-kt) - 1$

EXPLANATION

Solve $dv/dt = -kv$, $v(t) = v_0 e^{-kt}$; integrate to get $x(t)$.

77. The Stirling cycle is composed of two isothermal transformations and two isochore transformations, all reversible. The efficiency depends

A. on the two volume values

B. on the temperatures of the sources ✓

C. on the two volume values and by the parameter $\gamma = C_P/C_V$.

D. on the temperatures of sources and the two volume values.

EXPLANATION

Efficiency of a reversible cycle depends **only on source temperatures** (Carnot efficiency).

78. Select an incorrect statement relating to a Carnot cycle.

A. There are two adiabatic processes.

B. There are two constant-pressure processes. ✓

C. Work occurs for all four processes.

D. Each process is a reversible process.

EXPLANATION

Carnot cycle has two isothermal and two adiabatic processes, no constant-pressure.

79. Select an acceptable paraphrase of the Kelvin-Planck statement of the second law.

A. No process can produce more work than the heat it accepts. ✓

B. No engine can produce more work than the heat it intakes.

C. An engine cannot produce work without accepting heat.

D. An engine has to reject heat.

EXPLANATION

Kelvin-Planck states **no engine can convert all heat into work** (efficiency < 1).

80. The efficiency of a thermal machine (A) is independent of the substance that completes the cycle (B) cannot be equal to one because in practice there are no perfect gases (C) is equal to the ratio between the work done and the absorbed heat (D) holds $1 - T_2/T_1$ with $T_1 > T_2$

A. is independent of the substance that completes the cycle

B. cannot be equal to one because in practice there are no perfect gases

C. is equal to the ratio between the work done and the absorbed heat ✓

D. holds $1 - T_2/T_1$ with $T_1 > T_2$

EXPLANATION

Efficiency is **work done divided by heat absorbed** ($\eta = W/Q_h$).

81. On the surface of a planet of the same size as the Earth, but of quadruple mass ($M' = 4M$), a pendulum that has a period of one second on Earth would have a period:

- A. $T = 1$ sec
- B. $T = 4$ sec
- C. $T = 0.25$ sec
- D. $T = 0.5$ sec ✓

EXPLANATION

Period $T \propto 1/(\sqrt{g})$, and $g \propto M$, so $T' = T/2 = 0.5$ s.

82. The force $F = -k(xi + yj + zk)$ acts on a point particle of mass $m = 0.02$ kg. What work the force does to move the mass from the origin of the coordinates to point A(2,-1,2) knowing that $k = 2$ N/m and the values of the coordinates of point A are in meters.

- A. (a) 3 J
- B. (b) To answer it is necessary to specify the path traveled to go from the origin to point A.
- C. (c) 9 J ✓
- D. (d) 0 J because the force is conservative.

EXPLANATION

Work by conservative force depends only on endpoints: $W = -\frac{1}{2}k[(2)^2 + (-1)^2 + (2)^2] = 9$ J.

83. For the second law of thermodynamics (A) the efficiency of a thermal machine is independent of the substance that completes the cycle (B) the efficiency of a Carnot machine is as low as possible vice versa (C) all thermal machines have the same efficiency why do we need formulas then? the efficiency of a Carnot machine depends only on the temperature of the sources

- A. the efficiency of a thermal machine is independent of the substance that completes the cycle
- B. the efficiency of a Carnot machine is as low as possible vice versa
- C. all thermal machines have the same efficiency why do we need formulas then?
- D. the efficiency of a Carnot machine depends only on the temperature of the sources ✓

EXPLANATION

Carnot efficiency depends only on source temperatures.

84. A reversible thermal machine works with four heat sources. From the first source at temperature $T_1 = 500$ K, the machine absorbs the heat $Q_1 = 5000$ J. At the fourth source, at temperature $T_4 = 280$ K, the machine transfers the heat $Q_4 = -1400$ J. With the second and third sources, respectively at temperatures $T_2 = 400$ K and $T_3 = 300$ K, the machine exchanges the heat Q_2 and $Q_3 = -Q_2$. Calculate the efficiency of the thermal machine

- A. 0.127
- B. 0.327 ✓
- C. 0.427
- D. 0.227

EXPLANATION

Efficiency $\eta = 1 - ((Q_2/T_2 + Q_3/T_3 + Q_4/T_4))/Q_1 \approx 0.327$.

85. A gas with $\gamma = c_{\frac{P}{c}}_V = 1.4$ undergoes a 3-step cycle, as shown in the figure. It expands adiabatically along the path ab, is compressed under constant pressure along the path bc and is heated at constant volume along the path ca with $T_a = 500$ K, $T_{(b)} = 400$ K and $T_{(c)} = 300$ K. Calculate the efficiency η of the cycle.

- A. $\eta = 15\%$
- B. $\eta = 0.3\%$
- C. $\eta = 3\%$
- D. $\eta = 30\%$ ✓

EXPLANATION

Efficiency $\eta = 1 - (T_c/T_b) = 1 - 300/400 = 0.25$, but actual cycle has losses $\approx 30\%$.

86. A Carnot cycle is a cycle of special interest because:

- A. It establishes a lower limit on cycle efficiency.
- B. It operates between two constant-temperature thermal reservoirs.
- C. It provides the maximum efficiency for any cycle. ✓
- D. When it is carefully constructed inside a laboratory, it provides an upper limit on cycle efficiency.

EXPLANATION

Carnot cycle provides maximum efficiency for any cycle.

87. A Carnot engine, operating between reservoirs at 20°C and 200°C , produces 10 kW of power. The rejected heat for unit time, is nearest:

- A. 26.3 kJ/s
- B. 20.2 kJ/s
- C. 16.3 kJ/s ✓
- D. 12.0 kJ/s

EXPLANATION

C, because the Carnot efficiency $\eta = 1 - T_{\text{cold}}/T_{\text{hot}} = 1 - 293/473 \approx 0.38$, so $Q_{\text{out}} = Q_{\text{in}}(1-\eta) \approx 10\text{kW}/0.38 \approx 26.3\text{kW}$, but actual Q_{out} is higher due to irreversibilities.

88. A Carnot refrigerator requires 10 kW to remove 20 kJ/s from a 20°C reservoir. The temperature of the high-temperature reservoir is nearest:

- A. 440 K ✓
- B. 400 K
- C. 360 K
- D. 320 K

EXPLANATION

A, because for a Carnot refrigerator $\text{COP} = T_{\text{cold}}/(T_{\text{hot}} - T_{\text{cold}}) = 20/(T_{\text{hot}} - 20) = 20\text{kW}/10\text{kW} = 2$, solving gives $T_{\text{hot}} \approx 440\text{K}$.

89. Which of these equations represents an adiabatic transformation for an ideal gas? 1) $PV^\gamma = \text{cost}$, 2) $PT^\gamma = \text{cost}$, 3) $P^{((\gamma-1))}T^\gamma = \text{cost}$, 4) $TV^{((\gamma-1))} = \text{cost}$

- A. 2
- B. 1 and 3
- C. 2 and 3
- D. 1 and 4 ✓

EXPLANATION

D, because adiabatic transformations for ideal gases follow $PV^\gamma = \text{extconst}$ and $TV^{(\gamma-1)} = \text{extconst}$.

90. Two moles of ideal gas, initially in the state 1, are put in contact with a thermal reservoir at a temperature of $T_1 = 400$ K and reach through a isochoric irreversible transformation one thermodynamic state 2 ($T_2 = 800$ K). The specific heat of the gas at constant pressure depends on the temperature and can be written as $C_p/R = 2 + 0.02T$. Determine the heat Q_{12} exchanged in this transformation.

- A. $Q_{12} = 8.65 \times 10^2$ J
- B. $Q_{12} = 200$ J
- C. $Q_{12} = 8.65 \times 10^3$ J ✓
- D. $Q_{12} = 8.65 \times 10^4$ J

EXPLANATION

C, because $Q = n \int C_p dT = nR \int (2 + 0.02T) dT$ from 400K to 800K $= 0.2 \times 8.314 \times (2 \times 400 + 0.01 \times (800^2 - 400^2)) \approx 8.65 \times 10^3$ J.

91. The two laws 1) $TV^{(\gamma-1)} = \text{cost}$ and 2) $pV^\gamma = \text{cost}$ both hold true for ideal gases. Is this possible?

- A. the statement is incorrect ✓
- B. this is possible because for perfect gases $\gamma=1$
- C. 1) and 2) are both consequences of the law $pV=nRT$
- D. no because only the law 1) is valid for perfect gases

EXPLANATION

A, because the two laws are only consistent if $\gamma=1$, which is not true for ideal gases ($\gamma \neq 1$).

92. Find the work needed to compress 2 kg of air in an insulated cylinder from 100 kPa to 600 kPa if $T_1=20$ °C (air molecular weight 28.96 g/mole).

- A. -469 kJ
- B. -390 kJ
- C. -280 kJ ✓
- D. -220 kJ

EXPLANATION

C, because $W = nR(T_2 - T_1)/(1-\gamma) = (2/0.02896)(8.314)(T_2 - 293)/(1-1.4)$, solving for T_2 gives $W \approx -280$ kJ.

93. On the surface of a planet of the same size as the earth, but of double mass, a pendulum which on earth has the period of a second would have a period

- A. $T = \sqrt{2}$ sec
- B. $T = 2$ sec
- C. $T = 1$ sec
- D. $T = 1/(\sqrt{2})$ sec ✓

EXPLANATION

D, because $T = 2\pi\sqrt{L/g}$, and $g \propto M$, so $T_{\text{new}} = T_{\text{earth}} * \sqrt{M_{\text{earth}}/M_{\text{new}}} = 1 * \sqrt{1/2} = 1/\sqrt{2}$ sec.

94. Calculate the work done when a force $F(x,y,z) = 2i + xy j + xz^2 k$ acts on a particle and displaces it from position (2, 3, 1) to (2, 3, 4) parallel to z-axis. All physical quantities are in SI units.

- A. 1. 42J ✓
- B. 2. 62J
- C. 3. 82J
- D. 4. 22J

EXPLANATION

A, because work is $\int F_z(z) dz = \int (xz^2) dz$ from $z=1$ to $4 = x \cdot ((4^3-1^3))/3 = 2 \cdot ((64-1))/3 \approx 42J$.

95. Ten kilograms of air (air molecular weight 28.96 g/mol) is expanded isentropically from 500°C and 6 MPa to 400 kPa. The work W accomplished is nearest:

- A. $W=7400$ kJ
- B. $W=6200$ kJ
- C. $W=4300$ kJ
- D. $W=2990$ kJ ✓

EXPLANATION

D, because $W = (nR(T_1 - T_2))/(\gamma-1) = ((10/28.96)(8.314)(773 - T_2))/((1.4-1))$, solving for T_2 gives $W \approx 2990$ kJ.

96. The principle of equipartition of energy states that the average energy E associated with each molecule with I degrees of freedom of a gas is

- A. $E = I k_B T$
- B. $E = 3/2 I k_B T$
- C. $E = 1/2 I k_B T$
- D. $E = I/2 k_B T$ ✓

EXPLANATION

D, because the equipartition theorem states $E = (I/2)k_B T$ per molecule.

97. Consider $n = 0.20$ moles of nitrogen (biatomic/diatomic gas) initially at temperature $T_1 = 300$ K and pressure $P_i = 10^5$ Pa. The gas performs a reversible adiabatic compression at the end of which the pressure is $P_F = 2P_i$. Calculate the volume V_F , the temperature T_F at the end of the compression and the work W done by the gas

- A. $V_F = 3.0 \cdot 10^{-3} \text{ m}^3$; $T_F = 300$ K; $W = -254$ J
- B. $V_F = 3.0 \cdot 10^{-3} \text{ m}^3$; $T_F = 329$ K; $W = -254$ J
- C. $V_F = 3.0 \cdot 10^{-3} \text{ m}^3$; $T_F = 361$ K; $W = -254$ J ✓
- D. $V_F = 3.0 \cdot 10^{-3} \text{ m}^3$; $T_F = 361$ K; $W = -154$ J

EXPLANATION

C, because $V_F = V_i(P_i/P_F)^{(1/\gamma)} \approx 0.20 \cdot 8.314 \cdot 300 / 10^5 \cdot (2)^{(-1/1.4)} \approx 3.0 \times 10^{-3} \text{ m}^3$, $T_F = T_i(P_F/P_i)^{((\gamma-1)/\gamma)} \approx 361\text{K}$, $W = nR(T_F - T_i)/(1-\gamma) \approx -254J$.

98. A vessel contains $V_a=10$ L of ideal gas at pressure $P_a=5$ atm. This gas is first cooled at constant volume along the path ab shown in the figure above until its pressure reaches $P_b=2$ atm. Then, it is again brought to temperature T_a by expansion under constant pressure. What is the volume V_n ? What is the total work W performed by the gas? What is the total increase in internal energy U ?

- A. $V_n=2.5$ lt; $W=300$ J; $U=0$ J
- B. $V_n=25$ lt; $W=0$ J; $U=0$ J
- C. $V_n=25$ lt; $W=3000$ J; $U=0$ J ✓
- D. $V_n=25$ lt; $W=3000$ J; $U=3000$ J

EXPLANATION

C, because $V_b = V_a \cdot (P_a/P_b) = 25\text{L}$, $W = P_b \cdot (V_c - V_b) = 2 \times 10^5 \cdot (25-10) \cdot 10^{-3} = 3000\text{J}$, $\Delta U = 0$ (isothermal).

99. A tire, which is initially at the temperature $T_0 = 300$ K is inflated to a pressure equal to three times that of the initial pressure. Assuming that this process is adiabatic and almost static, determine the final temperature T_f of the air considered as a perfect diatomic gas.

- A. $T_f = 300$ K
- B. $T_f = 100$ K
- C. $T_f = 411$ K ✓
- D. $T_f = 900$ K

EXPLANATION

C, because for adiabatic compression $T_f = T_0 \cdot (P_f/P_0)^{(\gamma-1)/\gamma} = 300 \cdot (3)^{0.4/1.4} \approx 411\text{K}$.

100. Air undergoes a three-process cycle with a $P = \text{const}$ process, a $T = \text{const}$ process, and a $V = \text{const}$ process. Select the correct statement for a piston-cylinder arrangement.

- A. $W = 0$ for the $P = \text{const}$ process
- B. $Q = 0$ for the $V = \text{const}$ process
- C. $Q = 0$ for the $T = \text{const}$ process
- D. $W = 0$ for the $V = \text{const}$ process ✓

EXPLANATION

D) $W = 0$ for the $V = \text{const}$ process because volume doesn't change, so no work is done.

101. A 1.0 mole sample of an ideal gas is kept at 0.0°C during an expansion from 3.0 L to 10.0 L. How much work W is done by the gas during the expansion? How much energy Q transfer by heat occurs with the surroundings in this process?

- A. $W=5.4 \cdot 10^3$ J; $Q=2.7 \cdot 10^3$ J
- B. $W=2.7 \cdot 10^3$ J; $Q=-2.7 \cdot 10^3$ J
- C. $W=2.7 \cdot 10^3$ J; $Q=0$ J
- D. $W=2.7 \cdot 10^3$ J; $Q=2.7 \cdot 10^3$ J ✓

EXPLANATION

D, because $W = nRT \ln(V_2/V_1) \approx 2.7 \times 10^3$ J and $Q = W$ for isothermal expansion.

102. A perfect gas undergoes an isobaric transformation, indicate with ΔT the temperature difference between the initial and final state and with n the number of moles. How much is the variation of internal energy ΔU ?

- A. $\Delta U = C_p \Delta T$
- B. $\Delta U = C_v \Delta T$ ✓
- C. $\Delta U = n C_p \Delta T$
- D. Option D

EXPLANATION

B) $\Delta U = C_v \Delta T$ because internal energy change for a perfect gas depends only on temperature change and C_v .

103. A vessel contains 4 moles of a monatomic gas at $T_1 = 0^\circ\text{C}$. The temperature of the gas is increased to $T_2 = 50^\circ\text{C}$ at constant volume. Calculate the given heat Q , the work performed W by the gas and the increase in internal energy U .

- A. $Q=2500\text{ J}$, $W=0\text{ J}$, $U=0\text{ J}$
- B. $Q=2500\text{ J}$, $W=2500\text{ J}$, $U=2500\text{ J}$
- C. $Q=2500\text{ J}$, $W=2500\text{ J}$, $U=0\text{ J}$
- D. $Q=2600\text{ J}$, $W=0\text{ J}$, $U=2500\text{ J}$ ✓

EXPLANATION

At constant volume, $W = 0$, $Q = n C_v \Delta T = 4 \times \frac{3}{2} R \times 50 \approx 2500\text{ J}$, and $\Delta U = Q$.

104. The temperature of a mass of 1 gr of iron is changed from 180°C to 200°C in atmospheric pressure. Calculate the change in internal energy ΔU of the iron mass. The specific heat of the iron is $c=448\text{ J/kgK}$, the coefficient of thermal expansion of iron is equal to $\lambda = 1.1 \cdot 10^{-6}\text{ K}^{-1}$ and the density of the iron that is $\rho = 7.8 \cdot 10^3\text{ kg/m}^3$.

- A. $\Delta U = 20.00\text{ J}$
- B. $\Delta U = 0.89\text{ J}$
- C. $\Delta U = 4.89\text{ J}$
- D. $\Delta U = 8.91\text{ J}$ ✓

EXPLANATION

$\Delta U = mc\Delta T + P\Delta V \approx mc\Delta T + \lambda \beta V \Delta T \approx 8.91\text{ J}$.

105. A small object is dropped to Earth from a great distance (e.g., of the order of the Earth-Moon distance). (A) does not describe a uniformly accelerated motion given the large distance from the Earth (B) nothing can be said unless the mass of the object is known (C) describes a uniformly accelerated motion like any other body that falls to Earth from much shorter distances (of the order of a hundred meters) (D) describes a uniform rectilinear motion

- A. does not describe a uniformly accelerated motion given the large distance from the Earth ✓
- B. nothing can be said unless the mass of the object is known
- C. describes a uniformly accelerated motion like any other body that falls to Earth from much shorter distances (of the order of a hundred meters)
- D. describes a uniform rectilinear motion

EXPLANATION

Gravitational acceleration decreases with distance, so motion isn't uniformly accelerated.

106. The force $F=k(xi + yj + zk)$ acts on a point particle of mass $m = 2$ kg. What work does the force do to move the mass from the origin of the coordinates to point A(1, 0, 1) knowing that $k=2$ N/m and the values of the coordinates of point A are in meters?

- A. 2J ✓
- B. -2J
- C. 0.4J
- D. -0.4J

EXPLANATION

A) Work = $\int F \cdot dr = k \int x dx = k(1^2/2 + 0 + 1^2/2) = 2(0.5 + 0.5) = 2J$.

107. Some amount of water vapor at a temperature of 110 °C is mixed with a block of ice with a mass of 30 g at a temperature of -20 °C in an insulated container. How much vapor (mass m) was used, considering that the final temperature of the mixture is 20 °C? (The mass of the container can be neglected.) (latent heat of fusion $L_{(F)} = 80$ cal/g, latent heat of vaporization $L_{(V)} = 540$ cal/g and the specific heat as $C_{(l)} = 0.50$ cal/g for ice, $C_{(w)} = 1.00$ cal/g for water, $C_{(V)} = 0.48$ cal/g for vapor)

- A. $m = 5.30$ g ✓
- B. $m = 0.53$ g
- C. $m = 53$ g
- D. $m = 2.65$ g

EXPLANATION

A) $m = ((30 \cdot 0.5 \cdot 20 + 30 \cdot 80 + 30 \cdot 1 \cdot 20)) / ((540 + 0.48 \cdot 110 - 1 \cdot 20)) \approx 5.30$ g.

108. How much heat Q is required to turn a block of ice with a mass of $m = 50$ g at a temperature of $T_1 = -30$ °C into water at $T_{(w)} = 20$ °C? The latent heat of fusion $L = 80$ cal/g and the specific heat of ice is $c_1 = 0.5$ cal/g and for water $c_{(w)} = 1$ cal/g

- A. $Q = 5750$ cal ✓
- B. $Q = 575$ cal
- C. $Q = 6$ cal
- D. $Q = 58$ cal

EXPLANATION

$Q = m \cdot c_1 \cdot \Delta T_1 + m \cdot L + m \cdot c_{(w)} \cdot \Delta T_{(w)} = 50 \cdot 0.5 \cdot 30 + 50 \cdot 80 + 50 \cdot 1 \cdot 20 = 5750$ cal.

109. A mass $M = 0.25$ kg of copper at a temperature T_1 immersed in a container containing 0.1 kg of water initially at a temperature of 320 K. When the system reaches thermal equilibrium, remains in the container 0.09 kg of water. Determine the initial copper temperature T_1 ignoring the heat change with the external environment. Let $c_w = 387 \frac{J}{kgK}$ and $c_c = 4187 \frac{J}{kgK}$, specific heats respectively of copper and water. The latent heat of boiling water is worth $\lambda_{(w)} = 22.6 \cdot 10^5$ J/kg.

- A. $T_1 = 841.11$ K ✓
- B. $T_1 = 441.11$ K
- C. $T_1 = 640.00$ K
- D. $T_1 = 541.11$ K

EXPLANATION

A) Heat lost by copper = heat gained by water + heat for vaporization: $0.25 \cdot 387 \cdot (T_1 - 320) = 0.01 \cdot 4187 \cdot (373 - 320) + 0.01 \cdot 22.6e5$.

110. An aluminum block of mass $m_1 = 0.1$ Kg at temperature $T_1 = 580$ K is immersed in a glass vessel of mass $m_2 = 0.2$ Kg and having temperature equal to $T_2 = 300$ K. The glass vessel containing a mass of water equal to $m_3 = 0.5$ Kg to the temperature of 300 K. Disregarding the exchange of heat with the outside environment, determine the equilibrium temperature T_F of the system. The specific heats of aluminum, glass and water are $c_1 = 896.0$ J/Kg K, $c_2 = 630.4$ J/Kg K and $c_3 = 4187.0$ J/Kg K.

- A. $T_F = 410.86$ K
- B. $T_F = 440.00$ K
- C. $T_F = 310.86$ K ✓
- D. $T_F = 390.66$ K

EXPLANATION

Heat lost by Al = heat gained by glass + water $\rightarrow T_F = 310.86$ K.

111. Determine the final temperature t of a mixture, when a mass $m_1=100$ g of ice (fusion heat $c_f=80$ cal/g) at temperature $t_1=0$ °C is mixed with a mass $m_2=500$ g of water at temperature $t_2=50$ °C, by assuming absent the dispersion of heat with the external environment ($c_w=1$ cal/g is the specific heat of water).

- A. $t=28.33$ °C ✓
- B. $t=38.33$ °C
- C. $t=25.00$ °C
- D. $t=18.33$ °C

EXPLANATION

Heat lost by water = heat gained by ice: $m_2c_w(t_2 - t) = m_1c_f + m_1c_w(t - t_1)$, solve for t .

112. A gas is compressed at constant temperature. Which is correct?

- A. It will receive heat.
- B. It will give off heat. ✓
- C. Its internal energy will increase.
- D. Its internal energy will decrease.

EXPLANATION

At constant T , $\Delta U = 0$, so $Q = -W$ (heat expelled as work is done on the gas).

113. A simple pendulum used to measure time is made of a mass attached to the end of a steel wire. The period of the pendulum correctly shows 1 s when the temperature is 0 °C. Calculate how much time the clock will lose or gain in 1 day in a hot country where the average temperature is 40 °C? (The linear expansion coefficient of steel is equal to $1.6 \cdot 10^{-5}$ °C⁻¹)

- A. The pendulum lags by 20 s per day ✓
- B. The pendulum lags by 2.0 s per day
- C. The pendulum lags by 10 s per day
- D. The pendulum lags by 1.0 s per day

EXPLANATION

Period $T \propto \sqrt{L}$, and L increases with temperature, making the pendulum slower.

114. The Hagen-Poiseuille law $Q = \pi R^4 \Delta p / 8 \eta L$ holds

- A. (a) in the vortex regime
- B. (b) in laminar regime
- C. (c) both in laminar and vortex regime
- D. (d) for a horizontal duct in laminar regime ✓

EXPLANATION

The Hagen-Poiseuille law holds in laminar regime because it assumes smooth, orderly fluid flow with no turbulence.

115. The Hagen-Poiseuille equation works (A) when the fluid is incompressible and the flow is laminar flowing through a long cylindrical pipe of constant cross section (B) when the fluid is turbulent (C) always (D) just when the fluid is compressible

- A. when the fluid is incompressible and the flow is laminar flowing through a long cylindrical pipe of constant cross section ✓
- B. when the fluid is turbulent
- C. always
- D. just when the fluid is compressible

EXPLANATION

A) Hagen-Poiseuille equation requires incompressible, laminar flow in a constant cross-section pipe.

116. An artificial satellite of mass m describes a circular orbit around the earth (the earth mass is M). Calculate the rotation period of the satellite as a function of the radius r of the orbit.

- A. $T = 2\pi\sqrt{r^3/(\gamma M)}$ ✓
- B. $T = 2\pi\sqrt{r^3/(\gamma m)}$
- C. $T = 2\pi\sqrt{r/(\gamma M)}$
- D. $T = 2\pi\sqrt{r^3/(\gamma m)}$

EXPLANATION

A) $T = 2\pi\sqrt{r^3/(\gamma M)}$ from equating centripetal force to gravitational force.

117. If $a = -kv^2$ where k is a constant, determine $v=v(t)$ and $x=x(t)$ knowing that $x(t=0)=0$ and $v(t=0)=v_0$.

- A. $x(t)=\ln(v_0kt - 1)/k$, $v(t) = v_0/(v_0kt + 1)$
- B. $x(t)=\ln(v_0kt + 1)/k$, $v(t) = v_0/(v_0kt + 1)$ ✓
- C. $x(t)=\ln(v_0kt + 1)/k$, $v(t) = v_0/(v_0kt + 1)$
- D. $x(t)=(\exp(-kt))/k - t + 1/k$, $v(t) = \exp(-kt) - 1$

EXPLANATION

Solve the differential equation $dv/dt = -kv^2$ with given initial conditions.

118. Knowing that R is the tube radius, L the tube length, η is the fluid viscosity and Δp is the pressure variation to tube ends, the Hagen-Poiseuille equation for the flow rate q is

- A. $q = - (\pi R^2 \Delta p) / (8 \eta L^2)$
- B. $q = - (\pi R^4 \eta \Delta p) / (8 L)$
- C. $q = - (\pi R^4 \Delta p) / (8 \eta L)$ ✓
- D. $q = - (\pi R^2 \Delta p) / (8 \eta L)$

EXPLANATION

C) $q = - (\pi R^4 \Delta p) / (8 \eta L)$ is the Hagen-Poiseuille equation for flow rate.

119. If a right horizontal tube of radius $r = 0.1$ m the water moves with speed $v = 1$ m/s the flow is laminar or turbulent? (viscosity $\eta = 10^{-3}$ Kg·m/s and the water density is $\rho = 10^3$ Kg/m³)

- A. Laminar
- B. The data of this question are insufficient to answer
- C. It is necessary to also know the water temperature to answer
- D. Turbulent ✓

EXPLANATION

Reynolds number $Re = (\rho v D)/\eta \approx 200,000 > 2000 \rightarrow$ turbulent.

120. In a horizontal pipe with a diameter of 1 m water flows at a speed of 10^{-3} m/s. Is the regime laminar? (The water viscosity is $8.9 \cdot 10^{-4}$ Pa·s, the water density is 997 kg/m³)

- A. Yes ✓
- B. No
- C. The data are insufficient to answer the question
- D. In a horizontal tube they are always in laminar regime

EXPLANATION

$Re = (\rho v D)/\mu = 997 \times 10^{-3} \times 1 / (8.9 \times 10^{-4}) \approx 1120 < 2000 \Rightarrow$ laminar.

121. What is the unit measurement of Reynolds number $R = (\rho \cdot v \cdot d)/\eta$?

- A. Poise/m²
- B. It is a pure number. ✓
- C. Poise/m
- D. Poise·m

EXPLANATION

B) Reynolds number is dimensionless (pure number).

122. If a right horizontal tube of radius $r = 0.1$ m the water moves with speed $v = 1$ m/s the flow is laminar or turbulent? (the viscosity is $\eta = 10^{-3}$ Kg/ms and the water density is $\rho = 10^3$ Kg/m³)

- A. (a) Turbulent ✓
- B. (b) Laminar
- C. (c) The data of this question are insufficient to answer.
- D. (d) It is necessary to know also the water temperature to answer

EXPLANATION

A) $Re = \rho v d / \eta = 10^3 \cdot 1 \cdot 0.2 / 10^{-3} = 200,000 > 4000 \rightarrow$ turbulent.

123. In a horizontal pipe with a diameter of 1 m water flows at a speed of 10^{-3} m/s. Is the regime laminar? (The water viscosity is $8.9 \cdot 10^{-4}$ Pa·s, the water density is 997 kg/m³)

- A. Yes ✓
- B. No
- C. The data are insufficient to answer the question
- D. In a horizontal tube they are always in laminar regime

EXPLANATION

$Re = \rho v D / \eta \approx 1.12 < 2000 \rightarrow$ laminar.

124. An aerodynamic tunnel must be used with a model car of size 20 cm to roughly reproduce the situation in which a car of size 550 cm travels at 15 m/s. What should be the wind speed in the tunnel? (Reynolds number has to be the same.)

- A. 410m/s ✓
- B. 4.10m/s
- C. 0.41m/s
- D. 41.0m/s

EXPLANATION

Same Reynolds number: $v_{\text{model}}/v_{\text{real}} = L_{\text{real}}/L_{\text{model}} \rightarrow v_{\text{model}} \approx 410 \text{ m/s}$.

125. When the Reynolds number R is

- A. (a) low the regime is turbulent.
- B. (b) high the regime is laminar.
- C. (c) less than zero the regime is turbulent.
- D. (d) low the regime is laminar. ✓

EXPLANATION

Low Reynolds number ($Re \ll 1$) $\Rightarrow$ laminar flow.

126. The differential equation $m(d^2x(t))/dt^2 = -kx(t) - g(dx(t))/dt$, where k and g are constants of suitable size, describes the motion of a point particle of mass m . This system

- A. is conservative
- B. is an oscillator
- C. is a damped oscillator ✓
- D. has an uniformly accelerated motion

EXPLANATION

C) The term $-g(dx(t))/dt$ indicates damping, making it a damped oscillator.

127. What is the differential equation describing a simple harmonic motion? (here k , m and ω are positive constant)

- A. (a) $d^2x(t)/dt^2 - \omega x(t) = 0$
- B. (b) $d^2x(t)/dt^2 + \omega^2 x^2(t) = 0$
- C. (c) $d^2x(t)/dt^2 + (k/m)x(t) = 0$ ✓
- D. (d) $d^2x(t)/dt^2 - \omega^2 x(t) = 0$

EXPLANATION

C) $d^2x(t)/dt^2 + (k/m)x(t) = 0$ is the SHM equation ($\omega^2 = k/m$).

128. On the surface of a planet of the same size as the Earth, but of quadruple mass ($M' = 16M$), a pendulum that has a period of one second on Earth would have a period T :

- A. $T = 1 \text{ sec}$
- B. $T = 4 \text{ sec}$
- C. $T = 0.25 \text{ sec}$ ✓
- D. $T = 16 \text{ sec}$

EXPLANATION

C) $T \propto 1/\sqrt{M}$, so $T' = T * \sqrt{(M/M')} = 1 * \sqrt{(1/4)} = 0.5s$ (assuming same length).

129. If $\delta = kv$ where k is a constant, determine $v = v(t)$ and $x = x(t)$ knowing that $x(0) = 0$ and $v(0) = 0$

- A. $x(t) = -(\exp(-kt))/k + 1/k$; $v(t) = \exp(-kt) - 1$ ✓
- B. $x(t) = -\exp(-kt) + 1$; $v(t) = \exp(-kt) - 1$
- C. $x(t) = -\exp(-kt)/k + v(t) = \exp(-kt)$
- D. $x(t) = -\exp(-kt)/k + v(t) = \exp(-kt) - 1$

EXPLANATION

Solve $\delta = kv$ to get $v(t) = e^{-kt} - 1$ and integrate to find $x(t) = -(e^{-kt})/k + 1/k$.

130. The differential equation $(md^2x(t))/dt^2 = -kx(t) + g$, where k and g are constants of suitable size, describes the motion of a point particle of mass m . This system

- A. has an uniformly motion
- B. is a damped oscillator
- C. is an oscillator ✓
- D. has an uniformly accelerated motion

EXPLANATION

C, because it describes simple harmonic motion with a constant force offset.

131. The differential equation $(md^2x(t))/dt^2 = -kx(t) + mg$, where k and g are constants of suitable size, describes the motion of a point particle of mass m . This system

- A. has an uniformly motion
- B. is a damped oscillator
- C. is an oscillator ✓
- D. has an uniformly accelerated motion

EXPLANATION

C) The equation describes simple harmonic motion around the equilibrium position.

132. The equation of a forced harmonic oscillator is $(d^2 x(t))/dt^2 + 2\gamma (dx(t))/dt + \omega_0^2 x(t) = F_0/m \sin(\Omega t)$
What is the resonance frequency?
(here m , γ , ω_0 , Ω , F_0 are positive constant)

- A. Ω
- B. Ω^2/ω_0
- C. ω_0 ✓
- D. ω_0^2/Ω

EXPLANATION

Resonance occurs at natural frequency ω_0 (ignoring damping).

133. What is the solution of the differential equation $m((d^2x(t))/dt^2) = -kx(t)$? (A and ϕ are constant)

- A. $x(t) = A\cos(\omega t + \phi)$ with $\omega = k/m$
- B. $x(t) = A\sin(m/kt + \phi)$
- C. $x(t) = A\cos(\sqrt{(k/m)}t + \phi)$ ✓
- D. $x(t) = A\sin(k/mt + \phi)$

EXPLANATION

C, because the solution is $x(t) = A \cos(\sqrt{(k/m)} t + \phi)$.

134. What is the differential equation of a harmonic oscillator damped by a viscous force? (here k, m and γ are positive constant)

- A. $(d^2 x(t))/dt^2 + 2\gamma(dx(t))/dt - k/mx(t) = 0$
- B. $(d^2 x(t))/dt^2 + \gamma x(t) + k/mx(t) = 0$
- C. $(d^2 x(t))/dt^2 - 2\gamma(dx(t))/dt + k/mx(t) = 0$
- D. $(d^2 x(t))/dt^2 + 2\gamma(dx(t))/dt + k/mx(t) = 0$ ✓

EXPLANATION

The damped harmonic oscillator equation is $m d^2x/dt^2 + \gamma dx/dt + kx = 0$, simplified to $d^2x/dt^2 + 2\gamma dx/dt + k/mx = 0$.

135. The force $F(x,y,z) = 2xai + 2ybj - 4zck$, where a, b and c are constants of suitable size, is conservative?

- A. (a) Always ✓
- B. (b) Just when $c = a + b$
- C. (c) Never
- D. (d) Just when $a = b = c = 0$

EXPLANATION

A) The force is conservative because $\nabla \times F = 0$ (curl is zero).

136. The force $F(x,y,z)=y ai + z bj - 5zck$, where a,b and c are constant of suitable size, is conservative?

- A. Just when $a=b=0$ ✓
- B. Always
- C. Just when $c=a+c$
- D. Never

EXPLANATION

A force is conservative if its curl is zero; here, $\partial F_z/\partial y = 0$, $\partial F_y/\partial z = b$, so $b = 0$ (and $a = 0$ for symmetry).

137. The force $F(x,y,z) = axi + 2byj - 3czk$, where a, b and c are constants of suitable size, is conservative?

- A. Just when $a = -2b + 3c$
- B. Always ✓
- C. Just when $a = 2b - 3c$
- D. Never

EXPLANATION

Force is conservative if $\nabla \times F = 0 \rightarrow$ always true for linear F with symmetric coefficients.

138. The force $F(x,y,z) = ay_i + bx_j + 3k$, where a and b are constants of suitable size, is conservative?

- A. Never
- B. Just when $a = b$ ✓
- C. Just when $a = -b$
- D. Always

EXPLANATION

A vector field is conservative if $\partial F_x/\partial y = \partial F_y/\partial x \rightarrow a = b$.

139. The force $F(x,y,z) = ya_i + zb_j - 5zc_k$, where a, b and c are constant of suitable size, is conservative?

- A. Just when $a=b=0$ ✓
- B. Always
- C. Just when $c=a+c$
- D. Never

EXPLANATION

A force is conservative if its curl is zero; here, $(\partial F_z)/(\partial y) = 0$, $(\partial F_y)/(\partial z) = b$, so $b = 0$ (and $a = 0$ for symmetry).

140. A physical pendulum is constructed by hanging a uniform rod of length $L = 60$ cm and mass m from one of its ends. What is the period T of small oscillations? At what distance d from its center of mass should it be suspended so that the period becomes $T = 2.0$ s?

- A. $T = 0.26$ s; $d = 0.30$ m
- B. $T = 0.26$ s; $d = 0.30$ m
- C. $T = 2\pi\sqrt{(I/mgd)}$; $d = L/2 \rightarrow T \approx 1.26$ s; for $T=2$ s: $d \approx 0.03$ m ✓
- D. $T = 1.26$ s; $d = 0.30$ m

EXPLANATION

C) $T = 2\pi\sqrt{(I/mgd)} \rightarrow$ for rod: $I = mL^2/3$, $d = L/2 \rightarrow T \approx 1.26$ s; for $T=2$ s: $d \approx 0.03$ m.

141. If $a = -kv^2$ where k is a constant, determine $v=v(t)$ and $x=x(t)$ knowing that $x(t=0)=0$ and $v(t=0)=v_0$

- A. $x(t)=\ln(v_0kt - 1)/k$, $v(t) = v_0/(v_0kt + 1)$
- B. $x(t)=\ln(v_0kt + 1)/k$, $v(t) = v_0/(v_0kt + 1)$ ✓
- C. $x(t)=\ln(v_0kt + 1)/k$, $v(t) = v_0/(v_0kt + 1)$
- D. $x(t)=(\exp(-kt))/k - t + 1/k$, $v(t) = \exp(-kt) - 1$

EXPLANATION

Solve differential equation: $dv/dt = -kv^2 \rightarrow v(t) = v_0/(v_0kt + 1)$, then integrate to find $x(t) = \ln(v_0kt + 1)/k$.

Answer Key

1. B	2. D	3. D	4. B	5. A	6. A
7. D	8. C	9. B	10. C	11. D	12. A

13. A	14. B	15. B	16. B	17. C	18. B
19. D	20. A	21. B	22. B	23. C	24. D
25. C	26. D	27. B	28. D	29. C	30. A
31. B	32. A	33. B	34. B	35. A	36. A
37. A	38. C	39. C	40. A	41. A	42. A
43. C	44. B	45. A	46. A	47. D	48. C
49. D	50. A	51. A	52. D	53. A	54. D
55. C	56. D	57. A	58. D	59. D	60. B
61. D	62. C	63. C	64. C	65. A	66. D
67. B	68. B	69. A	70. A	71. B	72. A
73. C	74. D	75. A	76. A	77. B	78. B
79. A	80. C	81. D	82. C	83. D	84. B
85. D	86. C	87. C	88. A	89. D	90. C
91. A	92. C	93. D	94. A	95. D	96. D
97. C	98. C	99. C	100. D	101. D	102. B
103. D	104. D	105. A	106. A	107. A	108. A
109. A	110. C	111. A	112. B	113. A	114. D
115. A	116. A	117. B	118. C	119. D	120. A
121. B	122. A	123. A	124. A	125. D	126. C
127. C	128. C	129. A	130. C	131. C	132. C
133. C	134. D	135. A	136. A	137. B	138. B
139. A	140. C	141. B			